# Sharp Quantitative Faber–Krahn Stability via Route $\delta$ (Plan 2: a No-Selection, No-Graph, No-Schauder Proof)

Plan 2 / Route $\delta$: assembled document


**Abstract**

We give a proof of the sharp quantitative Faber–Krahn inequality

$$\Big(\frac{|\Omega|}{|B_1|}\Big)^{2/n}\big(\lambda_1(\Omega)-\lambda_1(B^*)\big)\ \geq\ c_{\mathrm{FK}}(n)\,\mathcal{A}(\Omega)^2,$$

with sharp exponent 2 on the Fraenkel asymmetry $\mathcal{A}$ (equivalently 1/2 on the scale-normalised spectral deficit) that is structurally distinct from the Brasco–De Philippis–Velichkov [11] proof and from the contradiction-free Plan 1 / Route A proof [8]. The present proof, which we call *Route* $\delta$, makes no use of any of the following four classical ingredients: (i) the selection / penalisation principle of Cicalese–Leonardi and BDV; (ii) the graph-entry / Hausdorff-to-$C^{1,\gamma}$ step driven by free-boundary regularity; (iii) Schauder bootstrap with Gagliardo–Nirenberg interpolation; (iv) the nearly-spherical Fuglede-type closure theorem. In their place it uses a finite-perimeter level-set deficit identity, a metric foliation $\rho \mapsto [F_\rho] \in \mathcal{X}$ in the $L^1$-quotient $\mathcal{X}$ of characteristic classes modulo translations, a centroid kinematic identity, a space–time radial-moment identity, and a slicing-then-squaring decomposition that converts an integrated quadratic boundary oscillation into a sharp metric trace. The proof is organised into seven self-contained building blocks (LevelSetIdentity, MetricFramework, CentroidBound, SpaceTimeIdentity, WeightedMetricTrace, BoundaryLayerTransfer, GlobalAssembly), each delivered as a separate LaTeX document. The last two stages of the chain (bounded reduction and the Kohler–Jobin rearrangement) are imported verbatim from the Plan 1 Final/ library, so that the methodological novelty is concentrated in the first five blocks.


# Contents

# 1 Introduction

## 1.1 The Faber–Krahn inequality

Let $\Omega \subset \mathbb{R}^n$, $n \geq 2$, be an open set of finite Lebesgue measure and let

$$\lambda_1(\Omega) \;=\; \min_{u \in H_0^1(\Omega)\setminus\{0\}} \frac{\int_\Omega |\nabla u|^2\,dx}{\int_\Omega u^2\,dx}$$

be the first Dirichlet eigenvalue of $-\Delta$ on $\Omega$. The *Faber–Krahn inequality* asserts that among open sets of a prescribed volume the ball uniquely minimises $\lambda_1$: writing $B^*$ for the ball with $|B^*| = |\Omega|$,

$$\lambda_1(\Omega) \;\geq\; \lambda_1(B^*), \tag{1}$$

with equality if and only if $\Omega$ is a ball (up to a set of capacity zero). This was conjectured by Lord Rayleigh in the planar case in the late nineteenth century and proved in general dimension by Faber [26] and Krahn [27] via Schwarz symmetrisation and the Pólya–Szegő principle; modern accounts appear in [28, 29, 30].

The natural way to measure the deviation from spherical symmetry is the *Fraenkel asymmetry*

$$\mathcal{A}(\Omega) \;=\; \inf_{x_0 \in \mathbb{R}^n} \frac{|\Omega\,\Delta\,(B^* + x_0)|}{|B^*|} \;\in\; [0, 2), \tag{2}$$

which vanishes precisely on translates of the ball. A quantitative Faber–Krahn inequality is an estimate of the form $(|\Omega|/|B_1|)^{2/n}(\lambda_1(\Omega) - \lambda_1(B^*)) \geq c\,\omega(\mathcal{A}(\Omega))$ for some modulus $\omega$. The ellipsoidal perturbation family $\Omega_\varepsilon = \{|x'|^2 + (1+\varepsilon)x_n^2 \leq 1\}$ shows $\mathcal{A}(\Omega_\varepsilon) \simeq \varepsilon$ and $\lambda_1(\Omega_\varepsilon) - \lambda_1(B^*) \simeq \varepsilon^2$, so the sharp modulus is $\mathcal{A}^2$.

## 1.2 Stability: prior work

The first systematic study of Faber–Krahn stability is due to Hansen–Nadirashvili [31] and Melas [32] for restricted classes of sets. Bhattacharya [33] reached every planar open set with exponent 3. Fusco–Maggi–Pratelli [13], using their sharp quantitative isoperimetric inequality [12], extended this to all dimensions with exponent 4 on $\mathcal{A}$.

The sharp conjecture with exponent 2 was settled in 2015 by Brasco–De Philippis–Velichkov [11]: there exists $\sigma(n) > 0$ with $|\Omega|^{2/n}\lambda_1(\Omega) - |B^*|^{2/n}\lambda_1(B^*) \geq \sigma\mathcal{A}(\Omega)^2$. The BDV strategy has two pillars: a Kohler–Jobin [19, 18] reduction of Faber–Krahn to the analogous sharp Saint–Venant stability, then a selection principle in the spirit of Cicalese–Leonardi [15] and Acerbi–Fusco–Morini [16] combined with free-boundary regularity [36] and a Fuglede-type expansion [37] to produce a nearly spherical contradiction. Subsequent work of Brasco–De Philippis [34] and Allen–Kriventsov–Neumayer [35] addresses related spectral stability inequalities via different routes.

A constructive-constant version of the BDV theorem was given in Plan 1 [8] (*Route A*): the BDV selection / contradiction is replaced by a quantitative *single-set* selection (run once on a single near-extremal set), the regularity package is applied to that single set, and the BDV nearly spherical theorem is invoked directly; no subsequence is taken. Plan 1 retains, however, all four of the BDV technical ingredients (selection, graph entry, Schauder, nearly-spherical closure); it removes only the contradiction.

The persistent open problem the present paper addresses is whether a proof of the sharp inequality can be given that uses *none* of the four ingredients above. The price for this is a new *finite-perimeter* infrastructure for the level-set foliation of the torsion function, which is the content of Plan 2 / Route $\delta$.

### 1.3 Plan 2's contribution

Plan 2 gives a proof of the sharp quantitative Faber–Krahn inequality in which the small-asymmetry chain (S)→(G)→(I)→(N) of Plan 1 is replaced by a *metric level-set chain.* Explicitly:

(a) *No selection principle.* No penalised functional is minimised. The original domain $\Omega$ is analysed as it stands; its volume-radius foliation $E_\rho = \{u_\Omega > t(\rho)\}$ is the structural object.

(b) *No graph entry.* No claim that $\partial\Omega$, or that any selected $\widetilde{\Omega}$, is a normal graph over a sphere of class $C^{1,\gamma}$ is made. All boundary integrals are on *reduced boundaries* $\partial^* E_\rho$ in the De Giorgi sense.

(c) *No Schauder interpolation.* No Kinderlehrer–Nirenberg hodograph straightening, no Lieberman oblique Schauder bootstrap, no Gagliardo–Nirenberg interpolation. The proof remains in $W^{2,p}_{\mathrm{loc}}$ throughout.

(d) *No nearly-spherical closure theorem.* BDV Theorem 3.3 and the Fuglede $H^{1/2}$ expansion are not invoked. The deficit-to-asymmetry conversion is purely metric.

In their place, the chain uses the following four new ingredients:

(1) *The metric foliation.* The torsion volume-radius parametrisation $E_\rho = \{u > t(\rho)\}$ with $|E_\rho| = \omega_n \rho^n$ defines a curve $\rho \mapsto [F_\rho]$ in the metric space $(\mathcal{X}, d_{\mathcal{X}})$ of $L^1$-classes modulo translations (Building Block MetricFramework). The Ambrosio–Gigli–Savaré metric framework [17] furnishes a metric derivative $|\dot{F}_\rho|_{\mathcal{X}}$, controlled by a Reshetnyak / coarea limit of the moving-region first variation.

(2) *The centroid kinematic identity.* The bulk centroid $\bar{z}_\rho = |E_\rho|^{-1} \int_{E_\rho} x\, dx$ satisfies

$$\bar{z}'_\rho \;=\; \frac{1}{|E_\rho|} \int_{\partial^* E_\rho} (x - \bar{z}_\rho)\, V_\rho \, d\mathcal{H}^{n-1},$$

with $V_\rho = -t'(\rho)/|\nabla u|$ the normal velocity. The right-hand side has *no homothety term*, so the bound on $\bar{z}'_\rho$ goes through the unconditional pair (Cauchy–Schwarz on $V_\rho$, FMP / Fusco–Julin at the centroid) and does not consume any per-$\rho$ radial-trace input (Building Block CentroidBound).

(3) *The space–time* $\Pi$ *identity.* The signed radial moment $\Pi(E_\rho, \bar{z}_\rho) = (n-1)\int (\mathbf{1}_{B_\rho(\bar{z}_\rho)} - \mathbf{1}_{E_\rho})|x - \bar{z}_\rho|^{-1} dx$ satisfies, by Fubini in $(x, \rho)$, $\int_{\rho_*}^{\rho_\delta} \Pi(E_\rho, \bar{z}_\rho)\, d\mu(\rho) = (n-1) \int_{\widetilde{\Omega}} \sigma(x) dx$ for an explicit Fubini density $\sigma$ (Building Block SpaceTimeIdentity).

(4) *Slicing-then-squaring.* The radial $L^1$ trace $T_1 = \int_{\partial^* E_\rho} \big|\, |x - \bar{z}_\rho|/\rho - 1 \big|\, d\mathcal{H}^{n-1}$ is split as $T_1 = T_1^{\mathrm{rad,slic}} + T_1^{\tan}$; the radial piece is bounded *linearly* via the good-ray equality of slice-wise volumes and a bad-ray multiplicity bound; the squared volume-Fraenkel $|E_\rho \Delta B_\rho|^2$ is controlled by $\Pi(E_\rho, \bar{z}_\rho)$ via a slice-rearrangement. Squaring before integrating preserves the sharp $\delta$-rate that the alternative $T_2$-route would leak at $\sqrt{\delta}$.

**Theorem 1.1** (Sharp quantitative Faber–Krahn, Plan 2 form)**.** *Let $n \geq 2$. There exists a constant $c_{\mathrm{FK}}(n) > 0$, depending only on the dimension, given explicitly by (8) below as a composition of named constants of the seven Plan 2 building blocks (and of the Plan 1* BoundedReduction *and* KohlerJobinTransfer *blocks at the dimensional radius $R_*(n)$), such that*

$$\Big(\frac{|\Omega|}{|B_1|}\Big)^{2/n}\big(\lambda_1(\Omega) - \lambda_1(B^*)\big) \;\geq\; c_{\mathrm{FK}}(n)\,\mathcal{A}(\Omega)^2$$

*for every open $\Omega \subset \mathbb{R}^n$ of finite positive measure. The exponent 2 on $\mathcal{A}$ is sharp.*

### 1.4 Organisation of the paper

The argument is organised as a chain of seven self-contained building blocks. Schematically,

LevelSetIdentity → MetricFramework → CentroidBound → SpaceTimeIdentity → WeightedM

The first six blocks are the geometric heart of Plan 2; the seventh imports the bounded reduction of [11, §5] (equivalently Plan 1's BoundedReduction) and the Kohler–Jobin transfer of Brasco [18] (Plan 1's KohlerJobinTransfer).

## 2 Notation and standing hypotheses

Fix $n \geq 2$ and $R \geq 2$. We use the following normalisations:

- $u_\Omega \in H^1_0(\Omega)$ is the variational torsion function: $-\Delta u_\Omega = 1$ in $\Omega$, $u_\Omega = 0$ on $\partial\Omega$.
- $T(\Omega) = \int_\Omega u_\Omega$ is the torsional rigidity, and $E(\Omega) = -T(\Omega)/2$ is the Saint–Venant energy.
- $\delta_T(\Omega) := E(\Omega) - E(B^*) \geq 0$ at fixed volume.
- $\lambda_1(\Omega)$ is the first Dirichlet eigenvalue of $-\Delta$.
- $\mathcal{A}(\Omega) = \inf_{x_0} |\Omega\Delta(B^* + x_0)|/|B^*| \in [0, 2)$.
- For $\rho \in [\rho_*, 1]$, $E_\rho = \{u_\Omega > t(\rho)\}$ with $|E_\rho| = \omega_n\rho^n$ and $\mu(d\rho) = (-t'(\rho))d\rho$.
- $\bar{z}_\rho = |E_\rho|^{-1}\int_{E_\rho} x\,dx$ is the centroid of $E_\rho$.
- $F_\rho = \rho^{-1}E_\rho$ is the rescaled level set; $[F_\rho] \in \mathcal{X}$ its class in $\mathcal{X} = \{L^1\text{-classes}\}/\mathbb{R}^n$ with $d_{\mathcal{X}}([A], [C]) = \inf_a |A\Delta(C + a)|$.
- $V_\rho(x) = -t'(\rho)/|\nabla u(x)|$, $H_{z,\rho}(x) = (x - z)\cdot\nu_{E_\rho}(x)/\rho$.
- $\rho_\delta := 1 - \kappa\sqrt{\delta_T(\Omega)}$; $\widehat{\rho} \in [\rho_\delta - C_*\delta_T, \rho_\delta]$ the metric-trace radius.
- $\omega_n = |B_1|$, $L_n = \lambda_1(B_1)$, $T_n = T(B_1) = \omega_n/(n(n+2))$.
- $R_*(n) = \max\{d(n), 2\}$ the BDV bounded-reduction radius; we fix $\rho_* = 1/2$.

## 3 The seven building blocks chained

### 3.1 Block I: LevelSetIdentity

**Theorem 3.1** (Exact deficit identity; [1])**.** *Let $\Omega \subset \mathbb{R}^n$ be open with $|\Omega| < \infty$. With $E_t = \{u > t\}$, $m(t) = |E_t|$, $\alpha(t) = \int_{\partial^* E_t} |\nabla u|^{-1} d\mathcal{H}^{n-1}$, $\beta(t) = \int_{\partial^* E_t} |\nabla u|\, d\mathcal{H}^{n-1}$, $D_H(t) = \alpha\beta - P(E_t)^2$, $D_I(t) = P(E_t)^2 - n^2\omega_n^{2/n} m(t)^{2-2/n}$,*

$$\boxed{\frac{2}{|\Omega|}\big(E(\Omega) - E(B)\big) = \frac{1}{|\Omega|}\int_0^{\|u\|_\infty} \frac{D_H(t) + D_I(t)}{n^2\omega_n^{2/n}\, m(t)^{1-2/n}}\, dt.} \tag{3}$$

*Both $D_H, D_I \geq 0$ a.e.; no boundedness or smoothness of $\Omega$ is needed. All integrals on the reduced boundary $\partial^* E_t$.*

### 3.2 Block II: MetricFramework

**Theorem 3.2** (Metric first variation; [2]). *For $\Omega \subset B_R$ with $|\Omega| = \omega_n$ and any Borel centre field $\{z_\rho\}$ with $|z_\rho| \leq R'$, there is $C_{n,\rho_*,R} > 0$ such that*

$$\boxed{|\dot{F}_\rho|_{\mathcal{X}} \leq C_{n,\rho_*,R} \inf_{a\in\mathbb{R}^n} \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho,\rho} - a \cdot \nu_\rho| \, d\mathcal{H}^{n-1}} \tag{4}$$

*for a.e. $\rho \in [\rho_*, 1]$.*

### 3.3 Block III: CentroidBound

**Theorem 3.3** ($H^1$-in-$\rho$ centroid bound (F); [3]). *$\rho \mapsto \bar{z}_\rho$ is Lipschitz on $[\rho_*, 1]$ and there is $C_{\mathrm{cent}}(n, R, \rho_*) > 0$ with*

$$\boxed{\int_{\rho_*}^{\rho_\delta} |\bar{z}'_\rho|^2 \, (-t'(\rho)) \, d\rho \;\leq\; C_{\mathrm{cent}} \, \delta_T(\Omega).} \tag{F}$$

*The proof uses the centroid kinematic identity $\bar{z}'_\rho = |E_\rho|^{-1} \int_{\partial^* E_\rho} (x - \bar{z}_\rho) V_\rho \, d\mathcal{H}^{n-1}$ (no homothety term), the unconditional $D_H$-Cauchy on $V_\rho$, FMP / Fusco–Julin at the centroid, and the integrated Talenti gap of Theorem 3.1.*

### 3.4 Block IV: SpaceTimeIdentity

**Theorem 3.4** (Space-time $\Pi$ identity and integrated bounds; [4]). *Conditional on* (F)*:*

$$\boxed{\int_{\rho_*}^{\rho_\delta} \Pi(E_\rho, \bar{z}_\rho) \, d\mu(\rho) = (n-1) \int_{\widetilde{\Omega}} \sigma(x) \, dx,} \tag{5}$$

*the integrated quadratic oscillation $\int\int |\nu - e_r|^2 \, d\mathcal{H}^{n-1} d\mu \leq C_{B^2} \delta_T$ ($B^2$-int), and the integrated radial trace $\int (\int |H_{\bar{z}_\rho,\rho} - 1| d\mathcal{H}^{n-1})^2 \, d\mu \leq C_{\mathrm{radial}} \delta_T$ ($T_1^2$-int). The proof uses $T_1 = T_1^{\mathrm{rad,slic}} + T_1^{\mathrm{tan}}$ with linear good-ray equality and the slice-rearrangement $|E_\rho \Delta B_\rho|^2 \leq C \, \Pi(E_\rho, \bar{z}_\rho)$.*

### 3.5 Block V: WeightedMetricTrace

**Theorem 3.5** (Weighted metric trace at $\widehat{\rho}$; [5]). *Conditional on* (F) *and ($B^2$-int), $\int (d_{\mathcal{X}}(F_\rho, B_1)^2 + |\dot{F}_\rho|^2_{\mathcal{X}}) d\mu \;\leq\; C \, \delta_T$ (integrated action), $\int |\dot{F}_\rho|^2_{\mathcal{X}} \, d\rho \;\leq\; C' \delta_T$ (kinetic bound), and there is $\widehat{\rho} \in [\rho_\delta - C_* \delta_T, \rho_\delta]$ with*

$$\boxed{d_{\mathcal{X}}(F_{\widehat{\rho}}, B_1)^2 \;\leq\; C_* \, \delta_T(\Omega).} \tag{6}$$

### 3.6 Block VI: BoundaryLayerTransfer

**Theorem 3.6** (Bounded sharp Saint–Venant; [6]). *There is $C(n, R, \rho_*) > 0$ such that every open $\Omega \subset B_R$ with $|\Omega| = \omega_n$ satisfies*

$$\boxed{\mathcal{A}(\Omega)^2 \leq C(n, R, \rho_*) \, \delta_T(\Omega).} \tag{7}$$

*The proof combines* (6) *with Lemma 8.3 of Theorem 3.1 ($\mathcal{A}(\Omega) \leq \mathcal{A}(E_{\widehat{\rho}}) + 2|\Omega \setminus E_{\widehat{\rho}}|/\omega_n$) and the squaring identity $(a + b)^2 \leq 2(a^2 + b^2)$; sharp exponent 2.*

### 3.7 Block VII: GlobalAssembly

**Theorem 3.7** (Sharp Faber–Krahn; [7]). *Combining* (7) *at the dimensional radius* $R_*(n)$ *with the BDV bounded reduction [9] and the Kohler–Jobin transfer [10], there is a dimensional constant* $c_{\mathrm{FK}}(n) > 0$ *given by*

$$c_{\mathrm{FK}}(n) := \min\left\{ \frac{4L_n\, c_{\mathrm{SV}}^{\mathrm{glob}}(n)}{(n+2)T_n},\ \frac{L_n(2^{2/(n+2)}-1)}{4} \right\}, \tag{8}$$

*such that for every open* $\Omega \subset \mathbb{R}^n$ *with* $0 < |\Omega| < \infty$,

$$\boxed{\left(\frac{|\Omega|}{|B_1|}\right)^{2/n}\big(\lambda_1(\Omega) - \lambda_1(B^*)\big) \geq c_{\mathrm{FK}}(n)\,\mathcal{A}(\Omega)^2.} \tag{9}$$

## 4 The main theorem and final assembly

**Theorem 4.1** (Sharp quantitative Faber–Krahn stability, Plan 2 form). *Let* $n \geq 2$. *There is a dimensional constant* $c_{\mathrm{FK}}(n) > 0$, *given by* (8), *such that for every open* $\Omega \subset \mathbb{R}^n$ *of finite positive measure,* $(|\Omega|/|B_1|)^{2/n}(\lambda_1(\Omega) - \lambda_1(B^*)) \geq c_{\mathrm{FK}}(n)\mathcal{A}(\Omega)^2$. *The exponent* 2 *on* $\mathcal{A}$ *is sharp.*

*Proof.* Fix $\rho_* = 1/2$. The chain Theorems 3.1–3.6 delivers the bounded Saint–Venant inequality at $R = R_*(n)$. Theorem 3.7 then internalises the $R$-dependence via the BDV bounded reduction and converts to Faber–Krahn via Kohler–Jobin. □

## 5 Constants chain

| Stage | Constant | Dependence |
|---|---|---|
| (L) deficit identity | — | — |
| (M) first variation | $C_{n,\rho_*,R} = \rho_*^{-n}$ | $n, \rho_*, R$ |
| (F) centroid bound | $C_{\mathrm{cent}}$ | $n, R, \rho_*$ |
| (Π) ($\mathrm{B}^2$-int) | $C_{B^2}$ | $n, R, \rho_*$ |
| (Π) ($\mathrm{T}_1^2$-int) | $C_{\mathrm{radial}}$ | $n, R, \rho_*$ |
| (K) action, kinetic, trace | $C, C', C_*$ | $n, R, \rho_*$ |
| (∂) bounded SV | $C(n, R, \rho_*)$ | $n, R, \rho_*$ |
| $\rho_* = 1/2$, $R \to R_*(n)$ | $C(n, R_*(n), 1/2)$ | $n$ |
| (FK) BDV constants | $C_9(n), \delta(n), d(n)$ | $n$ |
| (FK) global SV | $c_{\mathrm{SV}}^{\mathrm{glob}}(n)$ | $n$ |
| (FK) KJ multiplier | $4L_n/((n+2)T_n)$ | $n$ |
| (FK) final | $c_{\mathrm{FK}}(n)$ | $n$ |

**Verdict.** All $R$- and $\rho_*$-dependence is absorbed at the Block (∂) → (FK) interface; final constant $c_{\mathrm{FK}}(n)$ depends only on $n$.

## 6 Comparison with Plan 1 (Route A)

Both routes deliver Theorem 4.1. The structural difference is concentrated in the bounded sharp Saint–Venant stage. Plan 1 uses the chain (S) selection → (G) graph entry → (I) Schauder interpolation → (N) nearly-spherical closure ([8]). Plan 2 uses the chain (L) deficit identity → (M) metric first variation → (F) centroid bound → (Π) space-time + slicing-then-squaring → (K) metric trace → (∂) boundary-layer transfer.

| Tool | Plan 1 / Route A | Plan 2 / Route $\delta$ |
|---|---|---|
| Selection / penalised functional | uses | avoids |
| Graph entry ($C^{1,\gamma}$ normal graph) | uses | avoids |
| Schauder bootstrap (Lieberman) | uses | avoids |
| Gagliardo–Nirenberg interpolation | uses | avoids |
| Nearly-spherical closure (BDV Thm. 3.3) | uses | avoids |
| Fuglede $H^{1/2}$ expansion | uses | avoids |
| Free-boundary regularity (Alt–Caffarelli) | uses | avoids |
| BV coarea on $\partial^* E_t$ for a.e. $t$ | not used | uses |
| AGS metric framework, $\lvert\dot F_\rho\rvert_{\mathcal{X}}$ | not used | uses |
| Centroid kinematic identity (no homothety) | not used | uses |
| Space–time $\Pi$ identity | not used | uses |
| Slicing-then-squaring (good rays / bad rays) | not used | uses |
| Slice-rearrangement $\lvert E\Delta B\rvert^2 \le C\Pi$ | not used | uses |
| Sobolev–Sard (Figalli–Maggi App. A) | not used | uses |
| BDV bounded reduction [11, §5] | uses | uses |
| Kohler–Jobin transfer | uses | uses |

The methodological gain of Plan 2 is that no PDE regularity beyond $W^{2,p}_{\mathrm{loc}}$, no free-boundary regularity, no selection, no graph parametrisation, and no nearly-spherical closure is used. The price is the new infrastructure in Blocks (M)–(Π)–(K).

# 7 Open issues and audit trail

The Plan 2 chain has been audited in three waves on the markdown source side; each post-audit correction is reflected in the `WIP_X.tex` files.

**Wave 1** (`agent6-adversarial-audit.md`) flagged five issues: centre-choice ambiguity (Fraenkel vs. centroid); reliance on the open pointwise per-$\rho$ radial trace (R); missing Sobolev–Sard citation; broken Cauchy–Schwarz in (G4) §5.2; notational drift between centre choices.

**Wave 2** (`wave2-*`) addressed: the centroid choice $\bar z_\rho = |E_\rho|^{-1}\int_{E_\rho} x$; the integrated form ($\mathrm{B}^2$-int) replacing pointwise (R); the Figalli–Maggi App. A Sobolev–Sard citation; the kinetic bound (K) rederivation from ($\mathrm{B}^2$-int).

**Wave 3** (`wave3-F,G,H,J`) introduced two new structural inputs: (F) the $H^1$-in-$\rho$ centroid bound (CentroidBound); and the slicing-then-squaring decomposition (SpaceTimeIdentity §5). The latter replaces the broken Cauchy–Schwarz of (G4) §5.2 by the linear good-ray equality and the slice-rearrangement; squaring before integrating preserves the sharp $\delta$ rate.

**Status.** The Plan 2 proof closes in its integrated form only. Pointwise per-$\rho$ statements (R) and ($\mathrm{B}^2$) are not invoked. The integrated form, controlled by (F) of CentroidBound and ($\mathrm{B}^2$-int) of SpaceTimeIdentity, suffices for the metric-trace and boundary-layer-transfer chain. This is the unconditional honest content of Plan 2.

# References


[1] Plan 2 building block LevelSetIdentity, `WIP_LevelSetIdentity.tex`.

[2] Plan 2 building block MetricFramework, `WIP_MetricFramework.tex`.

[3] Plan 2 building block CentroidBound, `WIP_CentroidBound.tex`.

[4] Plan 2 building block SpaceTimeIdentity, `WIP_SpaceTimeIdentity.tex`.

[5] Plan 2 building block WeightedMetricTrace, `WIP_WeightedMetricTrace.tex`.

[6] Plan 2 building block BoundaryLayerTransfer, `WIP_BoundaryLayerTransfer.tex`.

[7] Plan 2 building block GlobalAssembly, `WIP_GlobalAssembly.tex`.

[8] Plan 1 master document, `Final/master.tex`.

[9] Plan 1 building block BoundedReduction, `Final/BoundedReduction.tex`.

[10] Plan 1 building block KohlerJobinTransfer, `Final/KohlerJobinTransfer.tex`.

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